

Chromatic Completeness and the Independence of Geometric Obstruction

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We establish a strict logical separation between two distinct phenomena in orthogonality hypergraphs: *chromatic completeness*, the possibility of assigning a single globally consistent nondegenerate spectrum to all contexts, and *geometric coordinatizability*, the existence of a faithful orthogonal representation by rays. A strong chromatic number larger than the Hilbert-space dimension obstructs only the former. It does not, by itself, obstruct the existence of a faithful orthogonal representation. We make this separation explicit by comparing two three-dimensional examples with the same strong chromatic number. A completed 25-ray version of the Yu–Oh configuration has strong chromatic number four and nevertheless possesses an explicit faithful orthogonal representation in $\mathbb{R}^3$. Conversely, Greechie’s G_{32} hypergraph also has strong chromatic number four, and has a separating and unital set of two-valued states, but we give an elementary algebraic proof that it admits no faithful orthogonal representation in $\mathbb{C}^3$. The obstruction in G_{32} is therefore not chromatic but projective-geometric: the incidence relations force two distinct atoms to collapse onto the same ray.

I. INTRODUCTION

An orthogonality hypergraph $H = (V, E)$ consists of a vertex set V and a collection E of hyperedges, called *contexts*. In the intended three-dimensional case each context is a triple of mutually orthogonal rank-one projectors, or equivalently a triple of mutually orthogonal rays. More generally, an n -uniform orthogonality hypergraph is intended to model contexts consisting of n mutually orthogonal rays in an n -dimensional Hilbert space.

A *faithful orthogonal representation* (FOR) of an n -uniform hypergraph H in $\mathbb{C}^n$ is an assignment

$$i \mapsto [v_i] \in \mathbb{P}(\mathbb{C}^n)$$

of distinct rays to distinct vertices such that every context $e = \{i_1, \dots, i_n\} \in E$ is represented by an orthonormal basis,

$$\langle v_{i_j}, v_{i_k} \rangle = 0 \quad (j \neq k).$$

Here and below the word faithful means, at minimum, that distinct vertices are assigned to distinct rays. The main negative result below uses only this minimal distinctness requirement. The existence of faithful orthogonal representations is a central question in the quantum logic of the Kochen–Specker theorem [1].

Recent work on chromatic contextuality [2, 3] emphasizes that if a hypergraph requires more than n colors, then it cannot support a representation in which every context is assigned the same fixed set of n spectral labels. This observation is correct, but it concerns an additional layer of structure: the global spectral labeling of an already specified incidence geometry. It does not settle whether the incidence structure itself can be coordinatized by rays.

The purpose of this paper is to separate these two issues. Chromatic completeness is a global labeling condition. Geometric coordinatizability is a projective-geometric realization condition. These two conditions are logically independent. The completed Yu–Oh configuration gives a coordinatizable hypergraph with strong chromatic number four in dimension three. Greechie’s G_{32} gives a hypergraph with the same strong chromatic number, and even a separating and unital set of two-valued states, but no faithful representation in $\mathbb{C}^3$. Thus the chromatic number, the supply of two-valued states, and geometric coordinatizability are three distinct layers of structure.

II. DEFINITIONS

The term “chromatic number” for hypergraphs is not completely uniform in the literature. Throughout this paper we use the following convention.

Definition 1 (Strong chromatic number). *Let $H = (V, E)$ be a hypergraph. A strong k -coloring is a map $c : V \rightarrow \{1, \dots, k\}$ such that any two vertices occurring in a common context have different colors. Equivalently, for every context $e \in E$, the restriction $c|_e$ is injective. The least such k is denoted by $\chi(H)$.*

For an n -uniform hypergraph, a strong n -coloring assigns all n colors to every context exactly once. This is the coloring notion relevant to nondegenerate spectral assignments.

Definition 2 (Chromatic completeness). *Let $H = (V, E)$ be an n -uniform orthogonality hypergraph and let ρ be a faithful orthogonal representation of H in $\mathbb{C}^n$. The representation ρ is chromatically complete if there exists a single set*

$$\Sigma = \{\lambda_1, \dots, \lambda_n\}$$

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of n distinct scalars and a map $c : V \rightarrow \Sigma$ such that for every context $e \in E$, the restriction $c|_e : e \rightarrow \Sigma$ is a bijection.

Thus each context receives the same spectrum, with each spectral value used exactly once.

Remark 3 (Chromatic completeness and strong coloring). *For an n -uniform hypergraph, chromatic completeness is exactly strong n -colorability. Indeed, if a faithful orthogonal representation is chromatically complete, the n spectral labels themselves define a strong n -coloring, because every context receives each label exactly once. Conversely, any strong n -coloring may be relabeled by an arbitrary fixed nondegenerate spectrum $\Sigma = \{\lambda_1, \dots, \lambda_n\}$, giving precisely the spectral assignment required in Definition 2.*

Consequently, if $\chi(H) > n$, no chromatically complete representation can exist. This is a statement about the possibility of a global spectral labeling. It is not, by itself, a statement about the existence of a ray representation.

III. SPECTRAL RELABELING

For a fixed geometric context, the numerical eigenvalues attached to a nondegenerate observable are not part of the ray geometry. They are labels of the rank-one spectral projections.

Lemma 4 (Spectral equivalence). *Let*

$$A = \sum_{i=1}^n a_i P_i$$

be a maximal observable on $\mathbb{C}^n$, where the spectral projections P_i are one-dimensional and the eigenvalues a_i are distinct. Then every maximal observable sharing the same spectral projections is of the form

$$B = f(A),$$

where f is injective on the spectrum of A . Conversely, every injective function on the spectrum of A produces a maximal observable with the same spectral projections.

Proof. Let

$$B = \sum_{i=1}^n b_i P_i$$

be another maximal observable with the same one-dimensional spectral projections. Define f on the finite spectrum of A by

$$f(a_i) = b_i, \quad i = 1, \dots, n.$$

Since B is maximal, the values b_i are pairwise distinct. Hence f is injective on $\{a_1, \dots, a_n\}$.

An explicit polynomial representative of f is given by Lagrange interpolation:

$$f(x) = \sum_{i=1}^n b_i \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - a_j}{a_i - a_j}.$$

The finite functional calculus then gives

$$f(A) = \sum_{i=1}^n f(a_i) P_i = \sum_{i=1}^n b_i P_i = B.$$

Conversely, if f is injective on the spectrum of A , then $\sum_i f(a_i) P_i$ has the same spectral projections as A and still has nondegenerate spectrum. $\square$

Therefore, at the level of a fixed projective measurement, changing the nondegenerate eigenvalues amounts to an injective relabeling of outcomes. Chromatic completeness asks whether such labels can be chosen globally and context-independently. Failure of chromatic completeness need not mean failure of the underlying ray geometry.

IV. INDEPENDENCE OF COLORING AND COORDINATIZATION

The following proposition makes the separation explicit. The examples are already available in dimension three.

Proposition 5 (Independence). *The strong chromatic number of an orthogonality hypergraph is neither a necessary nor a sufficient criterion for the existence of a faithful orthogonal representation. More explicitly, already for $n = 3$:*

- (i) *there are hypergraphs with $\chi(H) \leq 3$ that possess an FOR in $\mathbb{C}^3$, and hypergraphs with $\chi(H) \leq 3$ that possess no FOR in $\mathbb{C}^3$;*
- (ii) *there are hypergraphs with $\chi(H) > 3$ that possess a FOR in $\mathbb{R}^3$, and hypergraphs with $\chi(H) > 3$ that possess no FOR in $\mathbb{C}^3$.*

Proof. For $\chi(H) \leq 3$ with a FOR, take the single context

$$E = \{\{1, 2, 3\}\}.$$

It has $\chi(H) = 3$ and is represented by the standard basis of $\mathbb{C}^3$.

For $\chi(H) \leq 3$ without a faithful orthogonal representation, take

$$E = \{\{1, 2, 3\}, \{1, 2, 4\}\}.$$

This hypergraph has a strong 3-coloring, for instance

$$c(1) = 1, \quad c(2) = 2, \quad c(3) = c(4) = 3.$$

However, in any representation in $\mathbb{C}^3$, the first context forces $[v_3]$ to be the one-dimensional orthogonal complement of $\text{span}(v_1, v_2)$. The second context forces $[v_4]$ to be the same orthogonal complement. Hence $[v_3] = [v_4]$, contradicting faithfulness.

For $\chi(H) > 3$ with a FOR, use the completed Yu–Oh hypergraph described in Sec. V. It has $\chi = 4$ and an explicit faithful representation in $\mathbb{R}^3$.

For $\chi(H) > 3$ without a faithful orthogonal representation, use Greechie’s G_{32} hypergraph. Its strong chromatic number is four, as shown in Sec. VIB, while Theorem 6 proves that no faithful orthogonal representation in $\mathbb{C}^3$ exists. $\square$

V. A COMPLETED 25-RAY YU–OH HYPERGRAPH

The Yu–Oh construction starts from 13 rays in $\mathbb{R}^3$ [4] (for extensions see Refs. [5–7]). In one standard coordinatization these rays are

$$\begin{aligned} z_1 &= (1, 0, 0), & z_2 &= (0, 1, 0), & z_3 &= (0, 0, 1), \\ y_1^- &= (0, 1, -1), & y_2^- &= (1, 0, -1), & y_3^- &= (1, -1, 0), \\ y_1^+ &= (0, 1, 1), & y_2^+ &= (1, 0, 1), & y_3^+ &= (1, 1, 0), \\ h_0 &= (1, 1, 1), & h_1 &= (-1, 1, 1), & h_2 &= (1, -1, 1), \\ h_3 &= (1, 1, -1). \end{aligned}$$

The original Yu–Oh argument is not a parity proof based on the absence of all Kochen–Specker value assignments. Rather, KS value assignments to the 13 rays exist, but they fail to reproduce a state-independent quantum prediction expressed by the Yu–Oh inequality [4]. This is precisely why the example is useful here: it separates coordinatizability from stronger global consistency requirements.

We shall use the 3-uniform completion of the Yu–Oh orthogonality graph: for each orthogonal pair among the original 13 rays, a third ray is included so that the pair belongs to a full orthogonal triad. The resulting completed hypergraph has 25 vertices and 16 contexts. It is the completed version used in the chromatic-contextuality analysis of Ref. [3].

Let $H_{\text{YO}}^+ = (V_{\text{YO}}^+, E_{\text{YO}}^+)$, with $V_{\text{YO}}^+ = \{1, \dots, 25\}$, have contexts

$$\begin{aligned} E_{\text{YO}}^+ &= \{\{1, 2, 3\}, \{7, 8, 9\}, \{4, 5, 6\}, \{2, 8, 5\}, \\ &\quad \{3, 12, 25\}, \{9, 12, 16\}, \{6, 12, 17\}, \{3, 11, 14\}, \\ &\quad \{7, 11, 15\}, \{4, 11, 24\}, \{9, 10, 23\}, \{1, 10, 19\}, \\ &\quad \{4, 10, 18\}, \{1, 13, 20\}, \{7, 13, 21\}, \{6, 13, 22\}\}. \end{aligned} \quad (1)$$

The first four contexts represent the three $\{y_i^+, z_i, y_i^-\}$ triads and the coordinate triad $\{z_1, z_2, z_3\}$. The remaining contexts complete the orthogonal pairs involving h_0, h_1, h_2, h_3 .

An explicit coordinatization is obtained by assigning to vertex i the ray spanned by the following unnormalized vector $r_i \in \mathbb{R}^3$:

i	1	2	3	4	5	6	7	8	9	10	11	12	13
x_i	0	1	0	1	0	1	1	0	1	1	-1	1	1
y_i	1	0	1	1	0	-1	0	1	0	-1	1	1	1
z_i	1	0	-1	0	1	0	1	0	-1	1	1	1	-1

(2)

i	14	15	16	17	18	19	20	21	22	23	24	25
x_i	2	1	1	1	1	2	2	1	1	1	1	2
y_i	1	2	-2	1	-1	1	-1	-2	1	2	-1	-1
z_i	1	-1	1	-2	-2	-1	1	-1	2	1	2	-1

(3)

That is, $r_i = (x_i, y_i, z_i)^T$, and the represented ray is $[r_i]$. The first 13 vertices are labeled as follows:

$$\begin{aligned} r_1 &= y_1^+, & r_2 &= z_1, & r_3 &= y_1^-, \\ r_4 &= y_3^+, & r_5 &= z_3, & r_6 &= y_3^-, \\ r_7 &= y_2^+, & r_8 &= z_2, & r_9 &= y_2^-, \\ r_{10} &= h_2, & r_{11} &= h_1, & r_{12} &= h_0, & r_{13} &= h_3, \end{aligned}$$

where equality means equality of rays. The remaining twelve vertices are completion rays. For instance,

$$r_{14} \parallel r_3 \times r_{11}, \quad r_{15} \parallel r_7 \times r_{11}, \quad r_{23} \parallel r_9 \times r_{10},$$

and similarly for the other added vertices.

Direct evaluation of Euclidean inner products gives

$$r_i \cdot r_j = 0$$

whenever $i \neq j$ occur in a common context of E_{YO}^+ . Thus Eq. (2) gives a faithful orthogonal representation of H_{YO}^+ in $\mathbb{R}^3$.

The strong chromatic number of this completed hypergraph is four. A strong 3-coloring of H_{YO}^+ would restrict to a proper 3-coloring of the Yu–Oh 13-ray orthogonality graph, because every orthogonal pair among the original

13 rays appears in at least one completed triad. But the Yu–Oh orthogonality graph is not 3-colorable; the elementary case analysis is the one given in Ref. [3]. Briefly, after fixing the color of h_0 , the three vertices y_1^-, y_2^-, y_3^- must use only the two remaining colors. Up to symme-

try there are two cases. In both cases the forced colors on the coordinate vertices and on the adjacent y_i^+ vertices leave one of h_1, h_2, h_3 with no available color. Hence three colors do not suffice.

On the other hand, the following strong 4-coloring shows that four colors do suffice:

i	1	2	3	4	5	6	7	8	9	10	11	12	13
$c(i)$	0	2	1	0	3	1	0	1	2	1	2	0	2

(4)

i	14	15	16	17	18	19	20	21	22	23	24	25
$c(i)$	0	1	1	2	2	2	1	1	0	0	1	2

(5)

Each context in Eq. (1) contains three distinct colors. Therefore

$$\chi(H_{\text{YO}}^+) = 4 > 3,$$

although H_{YO}^+ has an explicit FOR in $\mathbb{R}^3$. This is the desired example showing that chromatic obstruction is not geometric non-coordinatizability.

VI. GREECHIE’S G_{32} HYPERGRAPH

We now turn to the converse phenomenon: a hypergraph with the same strong chromatic number as the completed Yu–Oh example, but with no faithful orthogonal representation in $\mathbb{C}^3$.

A. The incidence structure

Greechie’s diagram G_{32} [8] has vertex set $\{1, \dots, 15\}$ and ten blocks:

$$\begin{aligned} E(G_{32}) = \{ & \{1, 2, 3\}, \{3, 4, 5\}, \{5, 6, 7\}, \{7, 8, 9\}, \\ & \{9, 10, 11\}, \{11, 12, 1\}, \{4, 10, 13\}, \\ & \{6, 12, 14\}, \{8, 2, 15\}, \{13, 14, 15\} \}. \end{aligned} \quad (6)$$

Each block is intended to represent a triad of mutually orthogonal rays. The hypergraph has a separating and unital set of two-valued states, as shown in Ref. [2]; nevertheless, it is not 3-colorable. Thus it already separates the existence of two-valued states from chromatic completeness. The theorem below shows that it also separates chromatic obstruction from geometric obstruction.

B. The strong chromatic number of G_{32}

The hypergraph G_{32} has strong chromatic number four [2]. A strong 4-coloring is

i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$c(i)$	0	1	2	0	1	0	2	0	1	2	3	1	1	2	3

(7)

A direct check of Eq. (6) shows that each block contains three distinct colors.

To see that three colors do not suffice, form the dual incidence graph: its vertices are the ten blocks of G_{32} , and each atom of G_{32} is an edge joining the two blocks in which that atom occurs. Since every atom of G_{32} occurs in exactly two blocks, this dual is a cubic graph. For G_{32} the dual graph is the Petersen graph. A strong 3-coloring of the atoms of G_{32} would be precisely a proper 3-edge-coloring of this cubic dual graph: the three atoms incident with each block would have to receive three different colors. But the Petersen graph has chromatic index four, not three. Hence G_{32} has no strong 3-coloring, and therefore

$$\chi(G_{32}) = 4.$$

This agrees with the previous chromatic analysis of G_{32} in Ref. [2].

C. Non-coordinatizability theorem

Theorem 6. *The hypergraph G_{32} admits no faithful orthogonal representation in $\mathbb{C}^3$.*

Proof. We use the inner-product convention

$$\langle x, y \rangle = x^\dagger y,$$

so the inner product is conjugate-linear in the first argument and linear in the second.

Assume, for contradiction, that a faithful orthogonal representation exists. Since $\{13, 14, 15\}$ is a block, its

three rays form an orthonormal basis. After applying a global unitary transformation, we may fix

$$v_{13} = e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad v_{14} = e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad v_{15} = e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The three blocks meeting this central triangle force three pairs of vectors into coordinate two-planes. From $\{4, 10, 13\}$ we obtain

$$v_4 = \begin{pmatrix} 0 \\ a \\ b \end{pmatrix}, \quad v_{10} = \begin{pmatrix} 0 \\ -\bar{b} \\ \bar{a} \end{pmatrix}, \quad |a|^2 + |b|^2 = 1.$$

From $\{6, 12, 14\}$ we obtain

$$v_6 = \begin{pmatrix} c \\ 0 \\ d \end{pmatrix}, \quad v_{12} = \begin{pmatrix} -\bar{d} \\ 0 \\ \bar{c} \end{pmatrix}, \quad |c|^2 + |d|^2 = 1.$$

From $\{8, 2, 15\}$ we obtain

$$v_2 = \begin{pmatrix} p \\ q \\ 0 \end{pmatrix}, \quad v_8 = \begin{pmatrix} -\bar{q} \\ \bar{p} \\ 0 \end{pmatrix}, \quad |p|^2 + |q|^2 = 1.$$

These parametrizations are general: in each coordinate two-plane, once one normalized ray is chosen, the orthogonal ray is unique up to phase, and the chosen phases above lose no generality at the level of rays.

We shall use the following elementary projection identity. If $\{x, y, z\}$ is an orthonormal basis of $\mathbb{C}^3$, $u \perp x$, and $w \perp z$, then

$$\langle u, w \rangle = \langle u, y \rangle \langle y, w \rangle. \quad (8)$$

Indeed,

$$w = x \langle x, w \rangle + y \langle y, w \rangle + z \langle z, w \rangle.$$

Taking the inner product with u eliminates the first term because $u \perp x$, while the last coefficient vanishes because $w \perp z$. Only the middle term remains, proving Eq. (8).

Apply Eq. (8) to the block $\{1, 2, 3\}$, written as the orthonormal basis $\{v_3, v_2, v_1\}$. Since $v_4 \perp v_3$ by the block $\{3, 4, 5\}$ and $v_{12} \perp v_1$ by the block $\{11, 12, 1\}$, we get

$$\langle v_4, v_{12} \rangle = \langle v_4, v_2 \rangle \langle v_2, v_{12} \rangle. \quad (9)$$

With the parametrization above,

$$\langle v_4, v_{12} \rangle = \bar{b} \bar{c}, \quad \langle v_4, v_2 \rangle = \bar{a} q, \quad \langle v_2, v_{12} \rangle = -\bar{p} \bar{d}.$$

Thus

$$\bar{b} \bar{c} = -\bar{a} \bar{d} q \bar{p},$$

and after complex conjugation,

$$bc = -ad \bar{q} p. \quad (10)$$

Next apply Eq. (8) to the block $\{7, 8, 9\}$, written as the orthonormal basis $\{v_9, v_8, v_7\}$. Since $v_{10} \perp v_9$ by the block $\{9, 10, 11\}$ and $v_6 \perp v_7$ by the block $\{5, 6, 7\}$, we obtain

$$\langle v_{10}, v_6 \rangle = \langle v_{10}, v_8 \rangle \langle v_8, v_6 \rangle. \quad (11)$$

The explicit inner products are

$$\langle v_{10}, v_6 \rangle = ad, \quad \langle v_{10}, v_8 \rangle = -b \bar{p}, \quad \langle v_8, v_6 \rangle = -qc.$$

In the last equality the first component of v_8 is conjugated once more by $\langle x, y \rangle = x^\dagger y$; hence the factor is q , not $\bar{q}$. Therefore

$$ad = bc \bar{p} q. \quad (12)$$

Substituting Eq. (10) into Eq. (12) gives

$$ad = (-ad \bar{q} p) \bar{p} q = -ad |p|^2 |q|^2.$$

Hence

$$ad(1 + |p|^2 |q|^2) = 0.$$

The factor $1 + |p|^2 |q|^2$ is strictly positive, so

$$ad = 0.$$

If $a = 0$, then $|b| = 1$ and

$$v_4 = \begin{pmatrix} 0 \\ 0 \\ b \end{pmatrix} = be_3.$$

Thus $[v_4] = [v_{15}]$, contradicting faithfulness. If $d = 0$, then $|c| = 1$ and

$$v_6 = \begin{pmatrix} c \\ 0 \\ 0 \end{pmatrix} = ce_1.$$

Thus $[v_6] = [v_{13}]$, again contradicting faithfulness. Therefore no faithful orthogonal representation of G_{32} in $\mathbb{C}^3$ exists. $\square$

Remark 7. *The proof never uses the chromatic number of G_{32} . The contradiction comes from the propagation of transition amplitudes through the incidence geometry. The two factorizations forced by opposite parts of the outer cycle are algebraically incompatible with distinctness of the central-spoke rays. Thus the obstruction is geometric rather than chromatic.*

VII. CONCLUSION

The existence of a faithful orthogonal representation, the existence of a separating set of two-valued states, and the existence of a globally consistent spectral coloring

are distinct questions. The strong chromatic condition $\chi(H) > n$ rules out chromatic completeness in dimension n , because it rules out a global assignment of the same n eigenvalue labels to every context. It does not rule out a faithful orthogonal representation.

The completed Yu–Oh hypergraph makes this explicit: it has $\chi = 4 > 3$ but is faithfully represented by 25 rays in $\mathbb{R}^3$. Greechie’s G_{32} has the same strong chromatic number, and has a separating and unital set of two-valued states, but no faithful representation in $\mathbb{C}^3$. The obstruction in G_{32} is detected only by direct geometric analysis, not by coloring or by the mere presence or absence of two-valued states.

Thus chromatic contextuality concerns global consistency of spectral labels, whereas geometric coordinatizability concerns realizability of incidence relations by rays. These two layers of structure should not be con-

flated.

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